

WYNYARD, SK, Canada

WMO: 718650

Lat: 51.770N

Lon: 104.200W

Elev: 560

StdP: 94.77

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.5	-29.9	-36.7	0.1	-32.2	-33.8	0.2	-29.5	11.1	-8.8	10.1	-8.7	2.9	260	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	28.6	18.7	26.7	17.8	25.0	16.9	20.4	26.3	19.1	24.9	18.0	23.2	4.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	14.0	23.8	16.9	12.9	22.1	15.8	12.0	21.0	61.3	26.2	56.8	24.9	53.1	23.3	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.0	8.9	8.0	-36.2	32.3	3.1	2.2	-38.4	33.9	-40.2	35.2	-42.0	36.4	-44.2	38.0
			-36.2	22.7	3.0	1.5	-38.3	23.8	-40.0	24.7	-41.7	25.5	-43.8	26.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-15.1	-13.6	-6.9	2.8	10.1	15.2	17.8	16.7	11.9	3.9	-5.3	-12.4
	DBStd	13.38	8.93	8.13	7.94	6.01	4.97	3.66	3.16	3.48	4.71	5.24	7.02	8.19
	HDD10.0	3636	777	659	524	225	62	4	0	1	32	198	460	694
	HDD18.3	5968	1036	893	782	467	258	106	48	73	197	446	709	952
	CDD10.0	784	0	0	0	8	64	159	243	208	90	11	1	0
	CDD18.3	74	0	0	0	0	3	12	32	23	5	0	0	0
	CDH23.3	778	0	0	0	2	41	121	298	238	76	1	0	0
	CDH26.7	166	0	0	0	1	7	22	65	57	14	0	0	0
(14) Wind	WSAvg	3.8	3.9	3.8	4.0	4.1	4.1	3.6	3.3	3.2	3.7	3.9	4.0	3.8
(15) Precipitation	PrecAvg	449	20	13	16	29	42	84	74	65	42	31	20	19
	PrecMax	690	41	29	39	74	105	189	145	175	98	82	45	42
	PrecMin	256	4	3	2	6	3	17	33	8	3	0	2	5
	PrecStd	103	9	6	10	20	26	38	29	41	24	20	11	9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.4	4.1	11.2	21.1	27.3	29.4	30.9	31.3	28.8	21.9	13.1	5.5
		MCWB	2.0	1.7	6.0	11.3	14.6	18.3	21.4	19.4	16.5	13.0	7.5	2.5
	2%	DB	2.2	1.7	7.5	18.0	24.1	26.6	28.4	28.0	25.5	17.5	8.7	3.1
		MCWB	0.4	0.0	4.0	9.5	13.0	17.2	19.9	18.0	15.3	10.3	4.9	0.7
	5%	DB	0.0	-0.1	4.7	15.1	21.6	24.4	26.5	25.9	22.7	14.4	5.7	0.8
		MCWB	-1.4	-1.3	2.2	8.2	12.0	16.1	19.0	17.6	14.2	8.7	2.8	-0.7
	10%	DB	-2.5	-2.1	2.8	12.0	19.3	22.4	24.8	24.0	20.0	11.6	3.2	-1.4
		MCWB	-3.5	-3.1	1.0	6.5	11.0	14.8	17.9	16.5	13.0	7.4	1.2	-2.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	2.0	6.3	12.0	16.0	21.0	22.6	20.9	18.1	13.2	7.6	2.7
		MCDB	4.2	3.5	10.2	19.8	23.5	27.8	29.1	27.5	25.1	20.2	12.3	5.0
	2%	WB	0.4	0.2	4.1	10.2	14.2	18.5	21.1	19.5	16.3	11.0	5.0	0.8
		MCDB	2.1	1.5	7.2	16.8	21.9	24.1	26.8	25.9	23.3	15.6	8.3	2.8
	5%	WB	-1.4	-1.3	2.4	8.5	12.9	17.1	19.8	18.5	15.0	9.2	3.0	-0.7
		MCDB	-0.1	-0.1	4.7	14.0	20.4	22.6	25.4	24.5	21.5	13.6	5.4	0.9
	10%	WB	-3.5	-3.1	1.0	6.9	11.6	15.9	18.6	17.4	13.5	7.7	1.3	-2.5
		MCDB	-2.5	-2.1	2.6	11.6	17.8	21.0	23.6	22.6	19.2	11.3	3.1	-1.4
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	8.9	8.9	10.3	12.0	10.4	10.9	11.4	11.1	9.1	7.6	8.0
		MCDBR	9.5	9.1	9.9	14.9	15.1	13.7	13.8	14.1	15.3	13.6	10.3	9.2
	5% WB	MCWBR	8.1	7.9	7.2	8.4	7.2	6.6	7.2	6.8	7.4	7.9	7.1	7.7
		MCWBR	9.2	8.4	9.6	13.9	13.3	11.4	12.6	12.6	13.8	12.7	9.8	9.2
(40) Clear-Sky Solar Irradiance	taub	taub	0.242	0.288	0.287	0.321	0.328	0.348	0.358	0.362	0.313	0.279	0.249	0.255
		taud	2.311	2.173	2.356	2.419	2.450	2.429	2.398	2.384	2.512	2.558	2.477	2.246
	Ebn at Noon	Ebn at Noon	792	825	916	920	925	905	889	865	881	855	793	691
		Edh at Noon	61	95	96	103	105	109	110	106	83	65	52	55
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.19	2.26	3.75	4.86	5.85	5.96	6.23	5.06	3.70	2.15	1.08	0.84
	RadStd	RadStd	0.10	0.16	0.31	0.39	0.50	0.44	0.35	0.38	0.32	0.30	0.14	0.09

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page